

Meteors and the Milky Way, on a camera

Fragments of the Universe
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A one-page field card. Print it, fold it, take it out with you.

One setup, two subjects

One wide lens on a tripod collects the Milky Way and any meteor that crosses it. You are not choosing between the two subjects. Set up once, leave the camera running, and the same frames stack into a Milky Way image and catch whatever streaks through them on the way. From about April to September the core is up after dark.

SET IT MANUALLY, THEN LEAVE IT ALONE

Mode	Manual	Nothing should change between frames
File	RAW	You will lift shadows heavily later
Aperture	f/2.8, wide open	The meteor is instantaneous, so light gathered per instant is what counts
ISO	3200 dark site	1600 to 2500 suburban. High enough for faint trails, low enough to keep the sky from clipping
Shutter	20s at 14mm, 15s at 18mm	Long enough to be efficient, short enough that stars stay points
Focus	Manual, on a bright star	Magnify to 100x in live view and nail it, since the infinity mark is not reliable
White balance	Manual, around 4000K	Consistent colour across hundreds of frames
Long exposure NR	Off	It doubles your dead time, and every gap is a meteor missed
Stabilisation	Off	On a tripod it introduces drift rather than removing it
Drive	Continuous, remote locked on	Shoot without pause for hours

WHAT TO BRING

A camera that shoots manual and RAW, the widest and fastest lens you own, and a tripod that will not creep. A 14mm or 18mm at f/2.8 is close to ideal: wide enough to cover a large slice of sky, fast enough to record the meteor, and inexpensive in manual-focus form. The rest is small: a spare battery or two, a remote release that locks on, and a lens cloth for the dew.

WHERE TO POINT IT

Aim 40 to 60 degrees away from the radiant, and high up. Meteors near the radiant are foreshortened into short stubs, while the long photogenic streaks are the ones well off to one side. Anywhere high and dark works, so pick the direction with the least streetlight and the best foreground. In August, swinging south puts the Milky Way in the same frame.

THE PERSEIDS, AND STAYING LATE

The peak falls on the night of 12 to 13 August, on a new moon in 2026, which is as good as conditions get. Rates build through the night: early on the radiant is still low in the north-east, so much of the shower is happening below the horizon. If you can only stay out for part of the night, stay late rather than arriving early. The last two hours before dawn are usually the best of it.

LONGEST EXPOSURE BEFORE STARS STOP BEING POINTS

14mm	18mm	24mm	35mm	50mm
500 rule 36s	500 rule 28s	500 rule 21s	500 rule 14s	500 rule 10s
NPF rule 20s	NPF rule 15s	NPF rule 12s	NPF rule 8s	NPF rule 6s

Figures for a 24-megapixel full-frame body at f/2.8. The 500 rule dates from film and is generous for a modern sensor. The NPF rule accounts for aperture and pixel size and gives the stricter answer. You can happily sit between the two, since slightly trailed stars cost very little in a frame whose subject is a streak, and the longer shutter means fewer gaps between exposures. If the Milky Way is the priority for a given frame, stay closer to the NPF figure.

AFTERWARDS, FROM THE SAME SET OF FRAMES

The best meteor frame

Pick it and process it as a wide-field night shot.

A composite

Several meteors onto one sky, which is completely legitimate as long as you say that is what you did.

A star trail

Stack the whole run for a free second image. Stacking also cleans up the Milky Way, since averaging many frames is what removes noise.

A dark site changes everything

Meteors are forgiving and a bright Perseid will punch through a suburban sky. The Milky Way is not forgiving: under city glow the background is brighter than the galaxy, and no amount of processing brings back what the sensor never recorded. A dark sky also lets the fainter meteors register instead of vanishing. One decision, both subjects improved.

Compose, then stop looking at the camera

A frame of nothing but sky is a technical record. Put a tree line, a dune ridge or a person looking up along the bottom and it becomes a photograph. You are shooting hundreds of frames from a fixed position, so the composition you choose in the first five minutes is the one you keep. Then start the intervalometer, lie back and watch.

The full guide, with examples

Full walkthrough with example images, the reasoning behind each setting, and a phone version of this guide for nights when all you have is the one in your pocket.

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